CE205 Databases and Information Retrieval

Scenario Modelling and Database Implementation

Scenario: An electronics company manufactures products such as smart TVs, wireless speakers, and CCTV cameras. Each product is produced in batches and uses one or more electronic components. Components are supplied by suppliers.

The company also records customer orders. Each order is placed by one customer and contains only one type of product. For each order, the company stores the quantity ordered and unit price.

Currently, product, component, supplier, batch, customer, and order information is stored in a single spreadsheet, causing repeated data and update problems. The task is to identify the functional dependencies, normalise the relation to 3rd normal form, and implement it in MySQL.

GUIDELINE FOR SOLUTION

You will get ZERO marks if you start the normalisation process with multiple relations. You must begin normalisation with a single relation given in Appendix A (at the end of this assignment brief).

1. Functional Dependencies (FD) in the Relation [20%]

1.1 Identify FDs in the relation *R* (state assumption(s) for each FD, if any) **[15%]**
1.2 Show all FDs in a graphical representation (Remove redundant FDs, if any) **[5%]**

2. Primary Key of the Relation [10%]

2.1 Compute the primary key of the relation using the closure operator '+' **[5%]**
2.2 Find prime and non-prime attributes **[5%]**

3. Types of Functional Dependencies (FD) in the Relation [10%]

3.1 Determine partial FD(s) (if any) and give their justification **[5%]**
3.2 Determine transitive FD(s) (if any) and give their justification **[5%]**

4. Normalisation to 2NF [15%]

4.1 Decompose the relation from 1st normal form (1NF) to 2nd normal form (2NF) **[10%]**
4.2 Find primary key(s) and foreign key(s) in each decomposed sub-relations **[5%]**

5. Normalisation to 3NF [15%]

5.1 Decompose the sub-relation(s) from 2nd normal form (2NF) to 3rd normal form (3NF) **[10%]**
5.2 Find primary key(s) and foreign key(s) in each decomposed sub-relations **[5%]**
(List all decomposed relations clearly, including their keys)

6. Implementation using MySQL Workbench [30%]

7.1 Write code for each sub-relation in MySQL **[10%]**
7.2 Write code to insert data in each sub-relation **[5%]**
7.3 Write any of the following THREE queries in MySQL to retrieve information from the implemented data **[9%]** *(include screenshots in the report to show the output of each query **[6%]**)*

Queries

1. Show all products and their components.
2. Find products that use the Circuit Board.
3. Find all failed production batches.
4. Show complete order details with customer and product names.
5. Count how many components each product uses.
6. Find suppliers that supply more than one component.
7. Find the total spending by each customer.

MARKING SCHEME

The marks for each section of the assignment are specified next to them. However, if any section is completed incorrectly, it will affect the marks of all subsequent sections. For example, if functional dependencies (FDs) are not determined correctly, it will prevent the correct completion of all following sections.

SUBMISSION

- You will submit this coursework in **ONE PDF file** *(answers for Sections 1 to 5 and screenshots for queries)* and **THREE SQL files** *(one to create relations, one to insert data in relations, and one for queries)*
- The deadline to submit the assignment is **6 November 2026 (Check Faser for the submission deadline)**

Important Note:

Some randomly selected students will be asked to present the submitted work. If a student does not present, the student's total marks will be capped at 50%.

PLAGIARISM

You should work individually on this assignment. Anything you submit is assumed to be entirely your work. The usual University of Essex policy on plagiarism applies:

http://www1.essex.ac.uk/plagiarism/

Best of Luck!

Appendix A

Relation *R*

ProductCode	ProductName	ProductCategory	ComponentCode	ComponentName	SupplierID	SupplierName	BatchNo	ManufactureDate	InspectionResult	OrderNo	CustomerID	CustomerName	QuantityOrdered	UnitPrice
P101	Smart TV	Television	C201	LED Screen	S01	TechParts Ltd	B5001	2026-03-12	Pass	O9001	CU01	Digital World	10	320
P101	Smart TV	Television	C202	Circuit Board	S02	ElectroSupply	B5001	2026-03-12	Pass	O9001	CU01	Digital World	10	320
P102	Wireless Speaker	Audio	C202	Circuit Board	S02	ElectroSupply	B5002	2026-03-15	Pass	O9002	CU02	HomeTech Store	20	45
P102	Wireless Speaker	Audio	C203	Bluetooth Module	S02	ElectroSupply	B5002	2026-03-15	Pass	O9002	CU02	HomeTech Store	20	45
P101	Smart TV	Television	C201	LED Screen	S01	TechParts Ltd	B5003	2026-04-02	Pass	O9003	CU03	SecureTech Ltd	5	315
P101	Smart TV	Television	C202	Circuit Board	S02	ElectroSupply	B5003	2026-04-02	Pass	O9003	CU03	SecureTech Ltd	5	315
P103	CCTV Camera	Security	C202	Circuit Board	S02	ElectroSupply	B5004	2026-04-05	Fail	O9004	CU01	Digital World	8	75
P103	CCTV Camera	Security	C204	Camera Sensor	S03	Vision Components	B5004	2026-04-05	Fail	O9004	CU01	Digital World	8	75